

$$1. \text{Var}(3\text{-day}) = 3 \sigma_{\text{day}}^2 = 3(0.02)^2 = 0.0012$$

$$\sigma_{3\text{-day}} = \sqrt{0.0012} = 0.0346 = 3.46\%$$

$$2. \text{ Given } S_0 = 120, K = 115, T = 0.5, r = 0.03, C_{\text{market}} = 8.75$$

$$C(\sigma) = S_0 N(d_1) - K e^{-rT} N(d_2)$$

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}} \quad d_2 = d_1 - \sigma\sqrt{T}$$

Using Newton's method at $\sigma = 0.2$

$$\sigma_{n+1} = \sigma_n - \frac{C(\sigma_n) - 8.75}{V(\sigma_n)}, \text{ where } V = S_0 \phi(d_1)\sqrt{T} \text{ and}$$

$$\phi(d_1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right)$$

$$d_1 = \frac{\ln\left(\frac{120}{115}\right) + \left(0.03 + (0.2)^2 \times \frac{1}{2}\right) \times 0.5}{0.2 \times \sqrt{0.5}}$$

$$d_1 = 0.9477719$$

$$d_2 = 0.9477719 - 0.2 \times \sqrt{0.5} = 0.336297$$

$$N(d_1) = 0.683575 \quad N(d_2) = 0.631677$$

$$K e^{-rT} = 115 e^{-0.03 \times 0.5} = 113.2879$$

$$C = 120 \times 0.683575 - 113.2879 \times 0.631677 = 10.4677$$

$10.4677 > 8.57$, $\therefore$ at $\sigma = 20$, overprice

next:

$$10.4677 - 8.75 = 1.7177$$

$$\sigma_{n+1} = \sigma_n - \frac{C(\sigma_n) - 8.75}{V(\sigma_n)} \quad V = S_0 \phi(d_1) \sqrt{T}$$

$$V = 120 \times \phi(0.477719) \sqrt{0.5} = 30.20$$

$$\Delta \sigma \approx \frac{1.7177}{30.20} = 0.056875$$

$$0.2 - 0.056875 = 0.143125$$

Then I plugged in 0.143125 as new σ , repeated

2 more and for run 2 and 3, got

0.1412, meaning it's the correct vol.

Problem 3

$$\hat{\sigma}_t = 1.5\%, \quad S_{t-1} = 30, \quad S_t = 30.50, \quad \lambda = 0.94$$

$$\sigma_{t+1}^2 = \lambda \hat{\sigma}_t^2 + (1-\lambda) x_t^2$$

$$x_t = \ln\left(\frac{30.50}{30}\right) = 0.016529$$

$$x_t^2 = 0.00027322$$

$$\sigma_{t+1}^2 = 0.94(0.015)^2 + 0.06 \times 0.00027322$$

$$\sigma_{t+1}^2 = 0.00022789$$

$$\sigma_{t+1} = 0.015096 = \boxed{1.51\%} \quad \text{increase by } 0.01\%$$

4. Given

$$S_{t-\Delta} = 300 \quad S_t = 298, \quad \sigma_t = 1.3\% \quad \sigma_t^2 = 0.000169$$

$$x_t = \ln\left(\frac{298}{300}\right) = -0.006689, \quad x_t^2 = 0.000044743$$

a) given $\lambda = 0.94$

$$\begin{aligned} \hat{\sigma}_{t+\Delta}^2 &= \lambda \hat{\sigma}_t^2 + (1 - \lambda) x_t^2 = 0.94(1.69 \times 10^{-4}) + 0.06(4.4743 \times 10^{-5}) = \\ &= \boxed{1.2710\%} \end{aligned}$$

b) given $\alpha_0 = 2 \times 10^{-6}$, $\alpha_1 = 0.04$, $\beta_1 = 0.94$

$$\begin{aligned} \sigma_{t+\Delta}^2 &= \alpha_0 + \alpha_1 x_t^2 + \beta_1 \sigma_t^2 = 2 \times 10^{-6} + 1.7897 \times 10^{-6} + 1.58860 \times 10^{-4} = \\ &= \boxed{1.2753\%} \end{aligned}$$

5. Given:

$$\sigma_{t+\Delta}^2 = \alpha_0 + \alpha_1 X_t^2 + \beta_1 \sigma_t^2$$

$$\alpha_0 = 1.3465 \times 10^{-6}, \quad \alpha_1 = 0.083394, \quad \beta_1 = 0.910116$$

a)

In stat. equilibrium the uncond. var. doesn't move, $\therefore$

$$E[\sigma_{t+\Delta}^2] = E[\sigma_t^2] = V_L$$

and using tower

$$E[X_t^2] = E[E[X_t^2 | \mathcal{F}_{t-\Delta}]] = E[\sigma_t^2] = V_L$$

Rearranging formula $\alpha_0 = (1 - \alpha_1 - \beta_1) V_L$,

$$V_L = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}, \quad \text{meaning } \alpha_1 + \beta_1 < 1$$

$$\alpha_1 + \beta_1 = 0.993510 < 1 \quad \checkmark \quad 1 - 0.993510 = 0.00649$$

$$V_L = \frac{1.3465 \times 10^{-6}}{0.00649} = 2.07443 \times 10^{-4}$$

$$\sigma_L = \sqrt{2.07443 \times 10^{-4}} = 1.4404 \%$$

$$\sigma_L \sqrt{252} = 22.87 \% \quad \text{per annum}$$

$$b) E[X_t^2 | \mathcal{F}_{t-\Delta}] = \underbrace{Var(X_t | \mathcal{F}_{t-\Delta})}_{\sigma_t^2} + \underbrace{(E[X_t | \mathcal{F}_{t-\Delta}])^2}_0 = \sigma_t^2$$

Let $\rho = \alpha_1 + \beta_1$; $V_n = E[\sigma_{t+n\Delta}^2 | \mathcal{F}]$, so

$$V_{n+1} = \alpha_0 + \alpha_1 \underbrace{E[X_{t+n\Delta}^2 | \mathcal{F}]}_{V_n} + \beta_1 V_n = \alpha_0 + \rho V_n$$

$$V_L = \alpha_0 + \rho V_L$$

$$V_{n+1} - V_L = \rho(V_n - V_L) \Rightarrow E[\sigma_{t+n\Delta}^2] = V_L + \rho^n (\sigma_t^2 - V_L)$$

$$\sigma_t^2 = (0.01932)^2 = 2.99982 \times 10^{-4} \quad \sigma_t^2 - V_L = 9.25094 \times 10^{-5} > 0$$

Now at $n=10$

$$\rho^{10} = (0.99351)^{10} = 0.936963$$

$$E[\sigma_{t+10\Delta}^2] = 2.07473 \times 10^{-4} + 0.936963 (9.25094 \times 10^{-5}) = 2.94151 \times 10^{-4}$$

$$\sigma'_{10} = 1.7151\%$$

At $n=500$

$$\rho^{500} = 0.038559$$

$$2.07473 \times 10^{-4} + 0.038559 (9.25094 \times 10^{-5}) = 2.11040 \times 10^{-4}$$

$$\sigma'_{500} = 1.4527\%$$

$$c) E[X_t | \mathcal{F}_{t-\Delta}] = 0 \quad \text{Var}(X_t | \mathcal{F}_{t-\Delta}) = \sigma_t^2, \text{ since}$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$E[X_t^2 | \mathcal{F}_{t-\Delta}] = \text{Var}(X | \mathcal{F}_{t-\Delta}) + \overbrace{(E[X_t | \mathcal{F}_{t-\Delta}])^2}^0 = \sigma_t^2$$

Applying to GARCH,

$$V_{n+1} = E[\alpha_0 + \alpha_1 X_{t+n\Delta}^2 + \beta_1 \sigma_{t+n\Delta}^2 | \mathcal{F}], \text{ where}$$

expectation is linear and α_0 is const.

$$V_{n+1} = \alpha_0 + \alpha_1 E[X_{t+n\Delta}^2 | \mathcal{F}] + \beta_1 E[\sigma_{t+n\Delta}^2 | \mathcal{F}]$$

The last expectation is V_n by definition.

$$E[X_{t+n\Delta}^2 | \mathcal{F}] = E[E[X_{t+n\Delta}^2 | \mathcal{F}_{t+(n-1)\Delta}] | \mathcal{F}] = E[\sigma_{t+n\Delta}^2 | \mathcal{F}] = V_n$$

$$\text{So } V_{n+1} = \alpha_0 + \varphi V_n$$

Now

$$\alpha_0 + \varphi V_L = \alpha_0 + \varphi \cdot \frac{\alpha_0}{1-\varphi} = \frac{\alpha_0(1-\varphi) + \alpha_0\varphi}{1-\varphi} = \frac{\alpha_0}{1-\varphi} = V_L$$

because $\varphi \neq 1$

So

$$V_{n+1} - V_L = (\alpha_0 + \rho V_n) - (\alpha_0 + \rho V_L) = \rho(V_n - V_L)$$

$$V_n - V_L = \rho(V_{n-1} - V_L) = \rho^2(V_{n-2} - V_L) = \dots = \rho^n(V_n - V_L)$$

So

$$V_n = V_L + \rho^n (\overset{V_0}{\sigma_t^2} - V_L)$$

Lastly,

$$\alpha_1 + \beta_1 < 1 \quad \text{and} \quad \alpha_1, \beta_1 \geq 0 \Rightarrow 0 \leq \rho < 1, \therefore$$

$$\lim_{n \rightarrow \infty} \rho^n = 0$$

$$\begin{aligned} \lim_{n \rightarrow \infty} V_n &= \lim_{n \rightarrow \infty} [V_L + \rho^n (\sigma_t^2 - V_L)] = \\ &= V_L + \left(\lim_{n \rightarrow \infty} \rho^n \right) (\sigma_t^2 - V_L) = V_L \end{aligned}$$

So since $x \mapsto \sqrt{x}$ is continuous on $(0, \infty)$ and

$$V_L > 0$$

$$\lim_{n \rightarrow \infty} \sqrt{V_n} = \sqrt{\lim_{n \rightarrow \infty} V_n} = \sqrt{V_L} = \sigma_L$$